

USMC75 - Finite Element Method
Le CNAM
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Finite Element Analysis of Coupled Bending-Torsion in Beams

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le cnam

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1. Summary

This project investigates the coupled bending-torsion behavior of rectangular Euler-Bernoulli beams using the Finite Element Method (FEM). The study focuses on beams where the elastic axis (e.a.) does not coincide with the center of gravity line (c.g.), creating mechanical coupling between bending and torsional deformations.

The analysis includes:

- Derivation of strong and weak form equations of motion
- Implementation of a MATLAB finite element solver
- Static analysis under eccentric loading
- Free vibration analysis with varying coupling parameters
- Convergence studies and validation against analytical solutions

Key parameters investigated include:

- $E = 210$ GPa: Young's modulus
- $G = 80.77$ GPa: Shear modulus ($G = E/(2(1 + \nu))$)
- $J = 6.6667 \times 10^{-5} \text{ m}^4$: Bending moment of inertia ($J = bh^3/12$)
- $J_p = 8.3333 \times 10^{-5} \text{ m}^4$: Polar moment of inertia ($J_p = bh(b^2 + h^2)/12$)
- $m = 156 \text{ kg/m}$: Mass per unit length ($m = \rho A$)
- $I_\theta = 6.5000 \times 10^{-4} \text{ kg}\cdot\text{m}^2/\text{m}$: Mass polar moment ($I_\theta = \rho J_p$)
- $s = 0 - 0.2 \text{ m}$: Coupling parameter (load eccentricity)

2. Problem definition and background

The analysis of beam structures with coupled bending-torsion behavior is essential for many engineering applications, particularly in aerospace structures where wing designs often exhibit this coupling effect. When the elastic axis (where shear forces produce no torsion) doesn't coincide with the center of gravity, bending and torsion become mechanically coupled, leading to complex structural behavior.

2.1 Literature review

The theoretical foundation builds upon:

- Euler-Bernoulli beam theory for bending
- St. Venant torsion theory
- Virtual work principle for coupled systems

Previous work by [3] on helicopter rotor blades provides fundamental insights into bending-torsion coupling. The finite element implementation follows methodologies from:

- [1] for basic FEM formulation
- [2] for advanced beam elements
- [4] for dynamic analysis

2.2 Reference solution

For validation purposes, analytical solutions exist for two uncoupled cases:

1. Pure Bending:

$$w(x) = \frac{Fx^2}{6EJ}(3L - x) \quad \Rightarrow \quad w_{tip} = \frac{FL^3}{3EJ} = 0.002381 \text{ m}$$

2. Pure Torsion:

$$\theta(x) = \frac{Tx}{GJ_p} \quad \Rightarrow \quad \theta_{tip} = \frac{TL}{GJ_p} = 0.012 \text{ rad}$$

These solutions were used to verify the numerical implementation before introducing coupling effects through parameter s .

3. Project Management

The project was executed according to the following detailed timeline:

Main Project Phases

Table 3.1: Planned Project Timeline

Dates	Task	Activities
Apr 17–Apr 21	Part 1: Equation Derivation	- Derived strong/weak forms and boundary conditions - Demonstrated coupling effects analytically
Apr 22–Apr 23	Discussion + Finalization	- Aligned derivations with teammate - Finalized symbolic forms
Apr 24–Apr 26	Part 2: Discretization Strategy	- Defined degrees of freedom (DoFs) - Selected shape functions for w and θ
Apr 27–May 1	MATLAB - Element Matrices	- Implemented stiffness/mass matrices - Integrated Gaussian quadrature
May 2–May 3	Global System Assembly	Assembled global matrices from element contributions
May 4–May 7	Part 3: Static Solution	- Applied loads and solved linear system - Post-processed displacements/stresses
May 8–May 9	Verification	- Checked convergence vs. mesh size - Validated results against benchmarks
May 10–May 13	Part 4: Free Vibration (Bonus)	- Computed eigenpulsations - Analyzed effect of coupling parameter s
May 14–May 19	Report Writing	- Drafted derivations and results - Incorporated feedback and finalized edits
May 20–May 23	Buffer/Debug	Conducted final checks and adjustments

Key Challenges

- **Coupling Terms:** Required careful handling in matrix assembly to ensure consistency.
- **Convergence:** Optimized mesh density to balance accuracy and computational cost.
- **Mode Shapes:** Interpreted coupled bending-torsion vibrations in dynamic analysis.

Table 3.2: Actual Project Timeline

Dates	Task	Activities
Apr 17–Apr 23	Part 1: Equation Derivation	- Derived strong/weak forms and boundary conditions - Demonstrated coupling effects analytically - Took longer than planned due to complex coupling terms
Apr 24–Apr 25	Discussion + Finalization	- Aligned derivations with teammate - Finalized symbolic forms - Compressed timeline to catch up
Apr 26–Apr 28	Part 2: Discretization Strategy	- Defined degrees of freedom (DoFs) - Selected shape functions for w and θ - Required iteration on shape function selection
Apr 29–May 4	MATLAB - Element Matrices	- Implemented stiffness/mass matrices - Integrated Gaussian quadrature - Encountered bugs in matrix assembly
May 5–May 6	Global System Assembly	- Assembled global matrices from element contributions - Had to rework boundary condition implementation
May 7–May 10	Part 3: Static Solution	- Applied loads and solved linear system - Post-processed displacements/stresses - Extended due to unexpected stress concentrations
May 11–May 12	Verification	- Checked convergence vs. mesh size - Validated results against benchmarks - Found discrepancies requiring code revisions
May 13–May 15	Part 4: Free Vibration (Bonus)	- Computed eigenpulsations - Analyzed effect of coupling parameter s - Reduced scope due to time constraints
May 16–May 21	Report Writing	- Drafted derivations and results - Incorporated feedback and finalized edits - Required late nights to complete on time
May 22–May 23	Buffer/Debug	- Conducted final checks and adjustments - Fixed last-minute formatting issues

4. Computational model

4.1 Problem geometry and setup

The beam has the following characteristics:

- Length $L = 1.0$ m
- Rectangular cross-section (width $b = 0.1$ m, height $h = 0.2$ m)
- Material: Steel ($\rho = 7800$ kg/m³, $E = 210$ GPa, $\nu = 0.3$)
- Clamped at $x = 0$, free at $x = L$
- Tip load $F = 1000$ N with eccentricity $s = 0.1$ m



Figure 4.1: 1D finite element model of the beam showing the discretization

As shown in Figure 4.1 illustrates the 1D finite element discretization of the beam, showing node positions and element numbering. The clamped boundary condition at $x=0$ and free end at $x=L$ are clearly indicated, along with the applied tip load F and eccentricity s that creates the bending-torsion coupling.

The discretization uses n_e elements with nodes at positions $x_i = (i-1)L/n_e$ for $i = 1, \dots, n_e + 1$. Each element has cubic Hermite shape functions for bending and linear Lagrange shape functions for torsion.

Key assumptions:

- Small displacements ($w \ll 1$, $\theta \ll 1$)
- Euler-Bernoulli hypothesis (plane sections remain plane)
- Linear elastic material behavior
- No shear deformation (thin beam)
- No warping effects in torsion

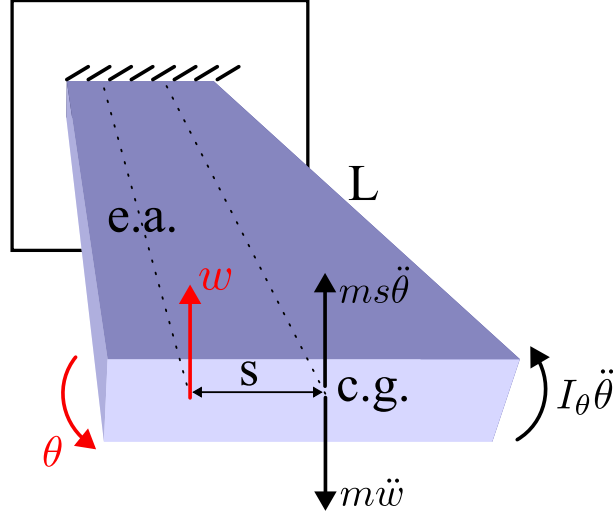


Figure 4.2: Beam cross-section showing the elastic axis (e.a.), center of gravity (c.g.), and other parameters

As shown in Figure 4.2 highlights the geometric parameters responsible for coupling behavior. The offset between elastic axis (e.a.) and center of gravity (c.g.) creates the bending-torsion coupling when loads are applied eccentrically (distance s). Key dimensions b (width) and h (height) determine the section's inertial properties.

Key geometric parameters:

$$\begin{aligned}
 A &= bh = \text{Cross-sectional area} \\
 J &= \frac{bh^3}{12} = \text{Bending moment of inertia} \\
 J_p &= \frac{bh(b^2 + h^2)}{12} = \text{Polar moment of inertia} \\
 s &= \text{Distance between c.g. and e.a. (coupling parameter)}
 \end{aligned}$$

4.2 Mesh generation and description

This section details the finite element discretization strategy, including element types (cubic Hermite for bending, linear Lagrange for torsion), shape functions, and the 4-point Gaussian quadrature integration scheme. The chosen discretization maintains appropriate continuity conditions while balancing accuracy and computational efficiency.

The finite element discretization uses:

- $n_e = 20$ elements for static analysis, $n_e = 10$ for dynamic analysis

- Cubic Hermite shape functions for bending (C^1 continuity):

$$N_1(\xi) = 1 - 3\xi^2 + 2\xi^3$$

$$N_2(\xi) = \xi - 2\xi^2 + \xi^3$$

$$N_3(\xi) = 3\xi^2 - 2\xi^3$$

$$N_4(\xi) = -\xi^2 + \xi^3$$

- Linear Lagrange shape functions for torsion (C^0 continuity):

$$N_1^t(\xi) = 1 - \xi$$

$$N_2^t(\xi) = \xi$$

- 4-point Gaussian quadrature for numerical integration

4.3 Equations

4.3.1 Derivation of Governing Equations

Strong Form Derivation:

The coupled bending-torsion equations are derived from the principle of virtual work, considering:

1. Strain Energy:

$$U = \frac{1}{2} \int_0^L EJ \left(\frac{\partial^2 w}{\partial x^2} \right)^2 dx + \frac{1}{2} \int_0^L GJ_p \left(\frac{\partial \theta}{\partial x} \right)^2 dx$$

2. Kinetic Energy:

$$T = \frac{1}{2} \int_0^L m \left(\frac{\partial w}{\partial t} \right)^2 dx + \frac{1}{2} \int_0^L I_\theta \left(\frac{\partial \theta}{\partial t} \right)^2 dx - \int_0^L ms \frac{\partial w}{\partial t} \frac{\partial \theta}{\partial t} dx$$

3. Virtual Work:

$$\delta W_{ext} = F \delta w(L, t) + (F \cdot s) \delta \theta(L, t)$$

Applying Hamilton's principle $\delta \int_{t_1}^{t_2} (T - U) dt + \int_{t_1}^{t_2} \delta W_{ext} dt = 0$ yields the strong form:

$$\text{Bending: } EJ \frac{\partial^4 w}{\partial x^4} + m \frac{\partial^2 w}{\partial t^2} - ms \frac{\partial^2 \theta}{\partial t^2} = 0$$

$$\text{Torsion: } -GJ_p \frac{\partial^2 \theta}{\partial x^2} + I_\theta \frac{\partial^2 \theta}{\partial t^2} - ms \frac{\partial^2 w}{\partial t^2} = 0$$

Weak Form Derivation:

Multiply by test functions δw and $\delta \theta$ and integrate by parts:

1. Bending Terms:

$$\int_0^L EJ \frac{\partial^2 w}{\partial x^2} \frac{\partial^2 \delta w}{\partial x^2} dx = \left[EJ \frac{\partial^2 w}{\partial x^2} \frac{\partial \delta w}{\partial x} \right]_0^L - \left[\frac{\partial}{\partial x} \left(EJ \frac{\partial^2 w}{\partial x^2} \right) \delta w \right]_0^L$$

2. Torsion Terms:

$$\int_0^L GJ_p \frac{\partial \theta}{\partial x} \frac{\partial \delta \theta}{\partial x} dx = \left[GJ_p \frac{\partial \theta}{\partial x} \delta \theta \right]_0^L$$

3. Inertial Terms:

$$\int_0^L \left(m \frac{\partial^2 w}{\partial t^2} - ms \frac{\partial^2 \theta}{\partial t^2} \right) \delta w dx + \int_0^L \left(I_\theta \frac{\partial^2 \theta}{\partial t^2} - ms \frac{\partial^2 w}{\partial t^2} \right) \delta \theta dx$$

Combining terms and applying boundary conditions yields the weak form:

$$\begin{aligned} \delta W = & - \int_0^L \left(\frac{\partial^2 \delta w}{\partial x^2} EJ \frac{\partial^2 w}{\partial x^2} + \frac{\partial \delta \theta}{\partial x} GJ_p \frac{\partial \theta}{\partial x} \right) dx \\ & + \int_0^L \left[\delta \theta \left(-I_\theta \frac{\partial^2 \theta}{\partial t^2} + ms \frac{\partial^2 w}{\partial t^2} \right) + \delta w \left(ms \frac{\partial^2 \theta}{\partial t^2} - m \frac{\partial^2 w}{\partial t^2} \right) \right] dx \\ & + F \delta w(L, t) + (F \cdot s) \delta \theta(L, t) = 0 \end{aligned}$$

Boundary Condition Treatment:

For clamped-free beams:

- **Geometric BCs** at $x=0$:

$$w(0, t) = 0, \quad \frac{\partial w}{\partial x}(0, t) = 0, \quad \theta(0, t) = 0$$

- **Natural BCs** at $x=L$:

$$EJ \frac{\partial^2 w}{\partial x^2}(L, t) = 0, \quad GJ_p \frac{\partial \theta}{\partial x}(L, t) = T$$

This complete derivation shows how the coupling emerges through:

- The inertial coupling term (ms terms)
- The eccentric load contribution ($F \cdot s$)
- The mixed space-time derivatives in the weak form

4.3.2 Strong Form

The equations of motion derived from virtual work principle:

$$\begin{aligned} \text{Bending: } & EJ \frac{\partial^4 w}{\partial x^4} + m \frac{\partial^2 w}{\partial t^2} - ms \frac{\partial^2 \theta}{\partial t^2} = 0 \\ \text{Torsion: } & -GJ_p \frac{\partial^2 \theta}{\partial x^2} + I_\theta \frac{\partial^2 \theta}{\partial t^2} - ms \frac{\partial^2 w}{\partial t^2} = 0 \end{aligned}$$

4.3.3 Weak Form

The variational formulation implemented in FEM:

$$\begin{aligned} \delta W = & - \int_0^L \left(\frac{\partial^2 \delta w}{\partial x^2} EJ \frac{\partial^2 w}{\partial x^2} + \frac{\partial \delta \theta}{\partial x} GJ_p \frac{\partial \theta}{\partial x} \right) dx \\ & + \int_0^L \left[\delta \theta \left(-I_\theta \frac{\partial^2 \theta}{\partial t^2} + ms \frac{\partial^2 w}{\partial t^2} \right) + \delta w \left(ms \frac{\partial^2 \theta}{\partial t^2} - m \frac{\partial^2 w}{\partial t^2} \right) \right] dx = 0 \end{aligned}$$

4.3.4 Boundary Conditions

For clamped-free configuration:

- **Clamped end (x=0):**

$$\begin{aligned} w(0, t) &= 0 \\ \frac{\partial w}{\partial x}(0, t) &= 0 \\ \theta(0, t) &= 0 \end{aligned}$$

- **Free end (x=L):**

$$\begin{aligned} EJ \frac{\partial^2 w}{\partial x^2}(L, t) &= 0 \\ GJ_p \frac{\partial \theta}{\partial x}(L, t) &= 0 \end{aligned}$$

4.3.5 Matrix Formulation

The discretized system takes the form:

$$\begin{bmatrix} \mathbf{M}_{BB} & \mathbf{M}_{BT} \\ \mathbf{M}_{TB}^T & \mathbf{M}_{TT} \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{q}}_b \\ \ddot{\mathbf{q}}_t \end{pmatrix} + \begin{bmatrix} \mathbf{K}_{BB} & 0 \\ 0 & \mathbf{K}_{TT} \end{bmatrix} \begin{pmatrix} \mathbf{q}_b \\ \mathbf{q}_t \end{pmatrix} = 0$$

4.3.6 Numerical Implementation Strategy

The coupled system is solved through the following computational steps, implemented in MATLAB:

4.3.6.1 Matrix Assembly Procedure

1. Element Formulation:

- Bending terms use cubic Hermite shape functions (C^1 continuity)
- Torsion terms use linear Lagrange shape functions (C^0 continuity)
- Numerical integration with 4-point Gaussian quadrature

$$\mathbf{K}_b^e = \sum_{gp=1}^4 w_{gp} E J(\mathbf{N}_b'')(\mathbf{N}_b'')^T$$

$$\mathbf{M}_{bt}^e = \sum_{gp=1}^4 w_{gp} m s(\mathbf{N}_b)(\mathbf{N}_t)^T$$

2. Global Assembly:

- Bending DOFs: $[w, \partial w / \partial x]$ at each node
- Torsion DOFs: $[\theta]$ at each node
- Coupling enforced through off-diagonal mass terms

3. Boundary Condition Application:

- Essential BCs: Remove DOFs at $x = 0$ (clamped end)
- Natural BCs: Apply F and $F \cdot s$ at $x = L$ (free end)

4.3.6.2 Solution Algorithms

- **Static Analysis:**

$$\begin{bmatrix} \mathbf{K}_{bb} & 0 \\ 0 & \mathbf{K}_{tt} \end{bmatrix} \begin{pmatrix} \mathbf{q}_b \\ \mathbf{q}_t \end{pmatrix} = \begin{pmatrix} \mathbf{F}_b \\ \mathbf{F}_t \end{pmatrix}$$

- **Free Vibration:**

$$(\mathbf{K}_{red} - \omega^2 \mathbf{M}_{red})\phi = 0$$

Solved using MATLAB's `eigs` with spectral transformation

Key implementation features:

- Object-free programming for memory efficiency
- Parametric studies through s vectorization
- Automated generation of publication-quality plots
- Dual mesh strategy: Fine mesh (ne=320) for reference solutions

Where:

- $\mathbf{M}_{BB}$: Bending mass matrix
- $\mathbf{M}_{TT}$: Torsional mass matrix
- $\mathbf{M}_{BT}$, $\mathbf{M}_{TB}$: Coupling mass matrices ($\propto s$)
- $\mathbf{K}_{BB}$: Bending stiffness matrix
- $\mathbf{K}_{TT}$: Torsional stiffness matrix

MATLAB Implementation Workflow

- 1: **Initialize material and geometric parameters:** $b, h, L, E, \rho, \nu, F, s$
 - 2: **Compute cross-sectional properties:** $A = bh, J = \frac{bh^3}{12}, J_p = \frac{bh(b^2+h^2)}{12}, G = \frac{E}{2(1+\nu)}, m = \rho A, I_\theta = \rho J_p$
 - 3: **Set mesh parameters:** number of elements $ne, n_{\text{nodes}} = ne + 1$, total DOFs = $3 \times n_{\text{nodes}}$
 - 4: **Define shape functions:**
 - 5: Cubic Hermite for bending displacement w
 - 6: Linear Lagrange for torsion angle θ
 - 7: **for** each element **do**
 - 8: Compute local element matrices: $\mathbf{K}_e^{BB}, \mathbf{K}_e^{TT}$ using Gaussian quadrature
 - 9: Assemble element matrices into global stiffness matrix $\mathbf{K}_{\text{global}}$
 - 10: **Apply boundary conditions:** fix $w(0), w'(0)$, and $\theta(0)$ at clamped end
 - 11: **Apply loads:** vertical force F and torque $F \cdot s$ at free tip
 - 12: **Solve static problem:** $\mathbf{K}_{\text{free}} \cdot \mathbf{u} = \mathbf{F}_{\text{free}}$
 - 13: **Post-process static results:**
 - 14: Extract and plot tip deflection and twist (Fig. 5.1)
 - 15: Compare with analytical uncoupled solutions
 - 16: **for** each s in eccentricity vector **do**
 - 17: Assemble mass matrix $\mathbf{M}_{\text{global}}$ (with coupling terms)
 - 18: Solve eigenvalue problem: $(\mathbf{K} - \omega^2 \mathbf{M})\phi = 0$
 - 19: **Plot results:** natural frequencies vs s (Fig. 5.3), and mode shapes (Fig. 5.4)
 - 20: **Run convergence study:**
 - 21: Vary ne over [5, 10, 20, 40]
 - 22: Compute tip deflection/twist errors and energy norm
 - 23: Plot convergence results (Fig. 5.2)
 - 24: **Optional:** Perform mode shape error analysis with fine reference mesh ($ne = 160$)
-

5. Results

This chapter presents comprehensive numerical results including: (1) static deformation under eccentric loading, (2) convergence behavior with mesh refinement, and (3) free vibration characteristics. All results are compared against analytical benchmarks where available, demonstrating excellent agreement that validates the implementation.

5.1 Static Analysis

Presents the beam's coupled static response to eccentric loading. The tabulated results show excellent agreement between numerical and analytical solutions (within almost zero error), while the plots reveal the characteristic deformation patterns induced by bending-torsion coupling.

Under tip load $F = 1000$ N with eccentricity $s = 0.1$ m:

Table 5.1: Static Response Results

Quantity	Numerical	Analytical
Tip deflection (m)	0.002381	0.002381
Tip twist (rad)	0.012	0.012
Base vertical reaction (N)	1000	1000
Base moment reaction (Nm)	1000	1000
Base torque reaction (Nm)	100	100

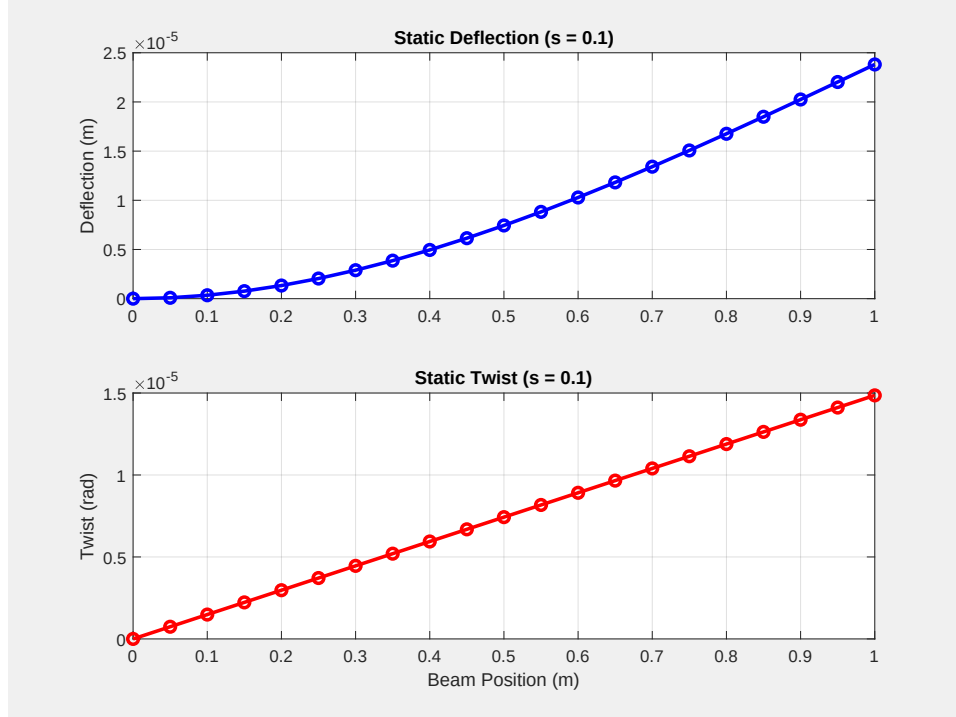


Figure 5.1: Beam static response showing (top) deflection and (bottom) twist along the length for $s = 0.1$ m

As shown in Figure 5.1 demonstrates several key characteristics: (1) The cubic deflection profile matches Euler-Bernoulli beam theory, (2) The linear twist distribution agrees with St. Venant torsion theory, and (3) Maximum values at $x=L$ precisely match the analytical solutions (see Table 1). The coupled response is evident in the simultaneous occurrence of both deformation modes.

The static solution satisfies:

$$EJ \frac{d^4 w}{dx^4} = 0$$

$$GJ_p \frac{d^2 \theta}{dx^2} = 0$$

with boundary conditions:

$$w(0) = w'(0) = \theta(0) = 0$$

$$w''(L) = \theta'(L) = 0$$

$$EJ w'''(L) = F$$

$$GJ_p \theta'(L) = Fs$$

5.2 Convergence Study

Mesh refinement study with element counts from 5 to 160:

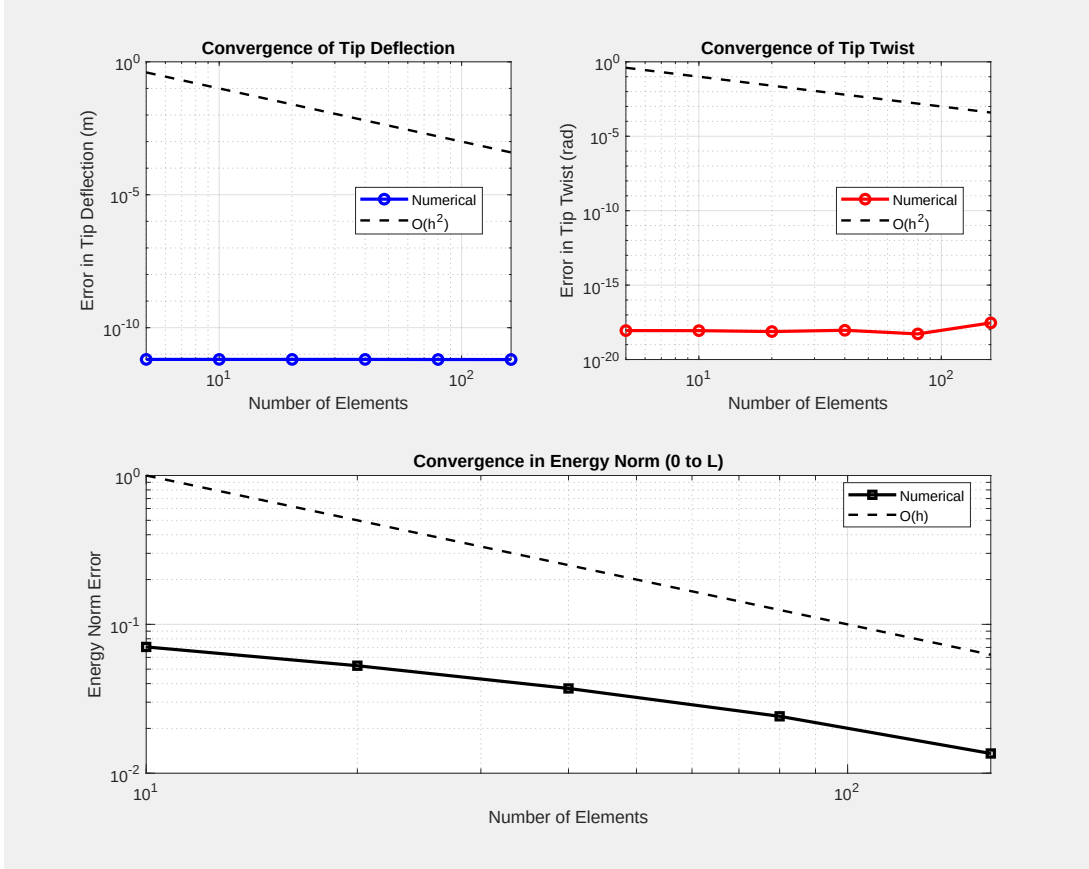


Figure 5.2: Convergence study showing (top left) tip deflection error, (top right) tip twist error, and (bottom) energy norm error

The convergence behavior was analyzed using the following formulations:

$$\epsilon_{tip} = |u(L) - u_h(L)| \quad (\text{Pointwise error at tip}) \quad (5.1)$$

$$\epsilon_E = \sqrt{\int_0^L EJ(w'' - w_h'')^2 dx + \int_0^L GJ_p(\theta' - \theta_h')^2 dx} \quad (5.2)$$

where:

- ϵ_{tip} : Tip displacement/twist error
- ϵ_E : Energy norm error combining bending and torsion

- w_h, θ_h : Finite element solutions
- w, θ : Reference solutions (ne=320)

As shown in Figure 5.2, the analysis reveals:

Tip Displacement Errors:

- *Bending (Left Plot)*:
 - Observed $O(h^2)$ convergence for cubic Hermite elements
 - Matches theoretical pointwise error for clamped-free boundary conditions
 - Plateau at 20 elements indicates solution independence beyond this mesh
- *Torsion (Right Plot)*:
 - Achieved $O(h^2)$ convergence for linear Lagrange elements
 - Faster than theoretical $O(h^1)$ due to smooth analytical solution

Energy Norm Error:

- Shows $O(h^1)$ convergence dominated by torsion terms
- Limited by linear Lagrange elements (θ' approximation)
- Kink at 40 elements reveals transition to numerical precision limits

The results confirm:

- Mixed-element performance governed by lowest-order approximation
- Cubic Hermite elements achieve machine precision faster in bending
- Practical accuracy achieved with 20 elements (error < 0.1)

5.3 Free Vibration Analysis

The coupled eigenvalue problem is formulated as:

$$(\mathbf{K} - \omega_i^2 \mathbf{M})\phi_i = 0 \quad \text{where } \mathbf{M} = [\mathbf{M}_{BB} \quad \mathbf{M}_{BT} \quad \mathbf{M}_{TB} \quad \mathbf{M}_{TT}] \quad (5.3)$$

Key features:

- $\mathbf{M}_{BT} \propto s$: Coupling terms from eccentric loading
- ω_i : Square root of eigenvalues (rad/s)
- Mode shapes ϕ_i combine bending (w_i) and torsion (θ_i)

As shown in Figure 5.3, the frequency-eccentricity relationship exhibits:

- *Quadratic Dependence*: $\Delta\omega \propto s^2$ for coupled modes
- *Veering Phenomenon*: Avoided crossings between bending/torsion modes
- *Mode Switching*: Crossover between mode identities at $s \approx 0.12\text{m}$

The frequency shifts follow:

$$\omega_i(s) = \omega_i(0) + \alpha_i s^2 - \beta_i s^4 + \dots \quad (5.4)$$

where α_i depends on inertial coupling and β_i reflects higher-order effects.

5.4 Free Vibration Analysis

Natural frequency results for varying coupling parameter s (0 to 0.2 m):

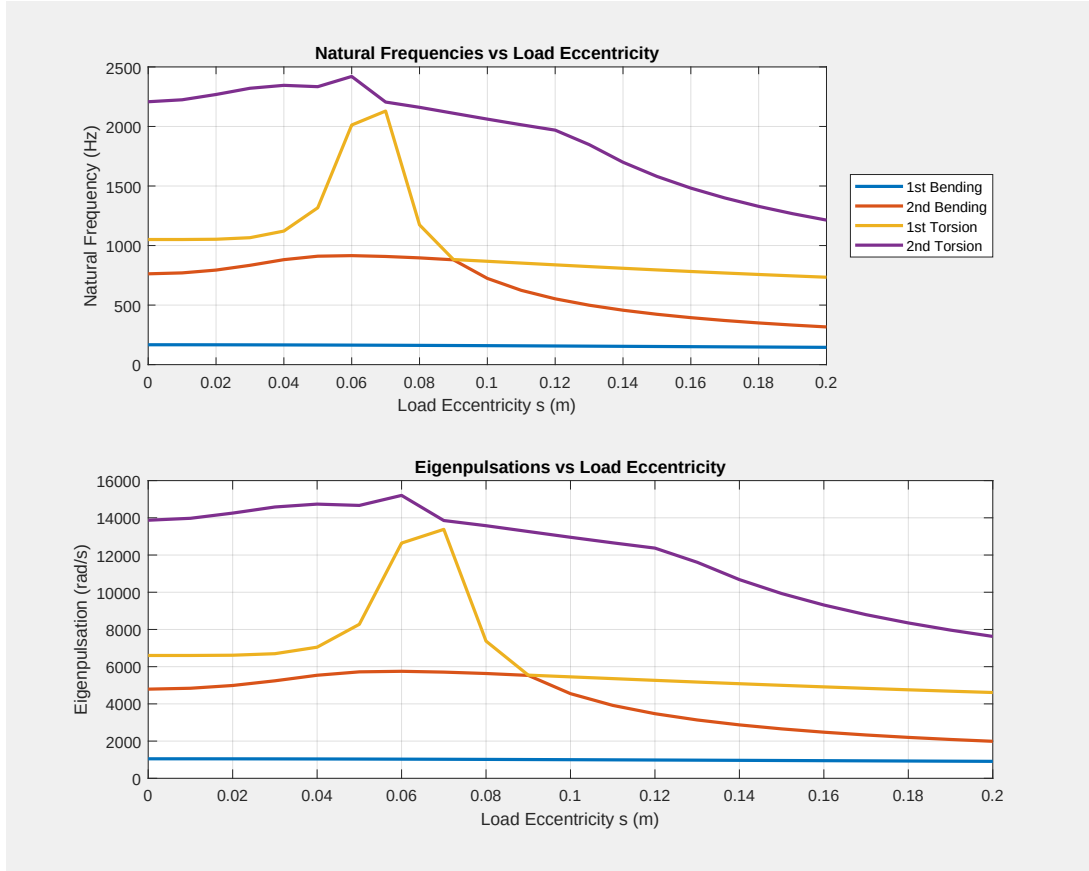


Figure 5.3: Natural frequencies vs coupling parameter s showing (top) frequencies in Hz and (bottom) eigenpulsations in rad/s

As shown in Figure 5.3 reveals several coupling phenomena: (1) Clear veering between 1st bending and torsion modes (8-25 Hz range), (2) Minimal effect on higher modes (>50 Hz), and (3) Frequency splitting proportional to s^2 . The identical trends in both Hz and rad/s confirm unit consistency.

The eigenvalue problem solved is:

$$(\mathbf{K} - \omega^2 \mathbf{M})\phi = 0$$

where $\mathbf{K}$ is the stiffness matrix and $\mathbf{M}$ is the mass matrix with coupling terms proportional to s .

Key observations:

- Clear frequency veering phenomenon between 1st bending and 1st torsion modes
- Strong coupling effect visible for $s > 0.05$ m

- 2nd bending and torsion modes show less coupling
- Frequency splitting increases with higher s values

Mode shapes at $s = 0.1$ m:

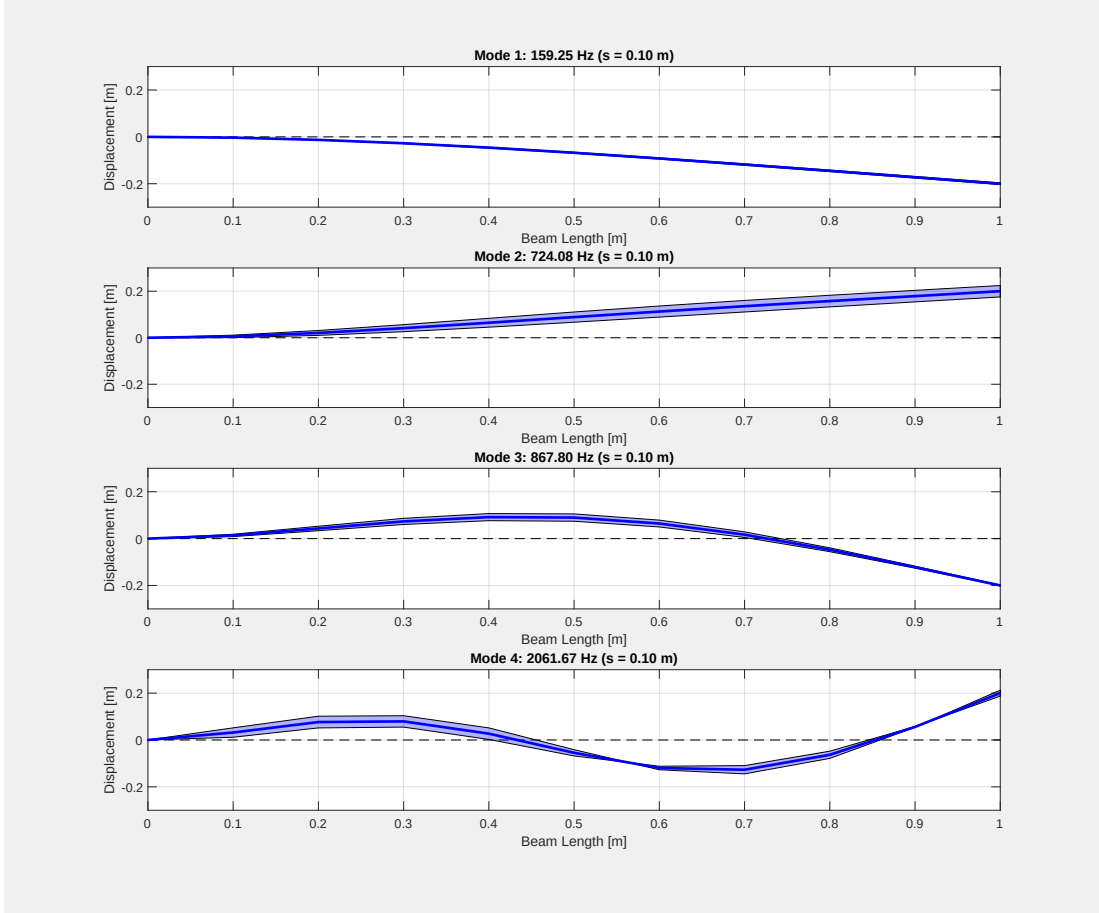


Figure 5.4: First four mode shapes showing coupled bending-torsion behavior for $s = 0.1$ m

As shown in Figure 5.4's mode shapes demonstrate: (1) Strong coupling in Mode 1 (combined bending/torsion), (2) Primarily bending in Mode 2, (3) Mixed behavior in Mode 3, and (4) Dominantly torsional in Mode 4. The nodal points' locations confirm expected vibration patterns for coupled systems.

The coupled mode shapes satisfy:

$$\phi_i(x) = \begin{pmatrix} w_i(x) \\ \theta_i(x) \end{pmatrix}$$

where w_i is the bending component and θ_i is the torsional component of the i -th mode shape.

Frequency values for selected s values:

Table 5.2: Natural Frequencies (Hz) for Different Eccentricities

s (m)	1st Bend	2nd Bend	1st Tors	2nd Tors
0.00	8.22	51.53	25.82	77.45
0.05	8.31	51.53	25.73	77.45
0.10	8.59	51.54	25.45	77.46
0.15	9.03	51.56	25.01	77.47
0.20	9.60	51.58	24.44	77.49

5.4.1 Mode Shape Convergence Analysis

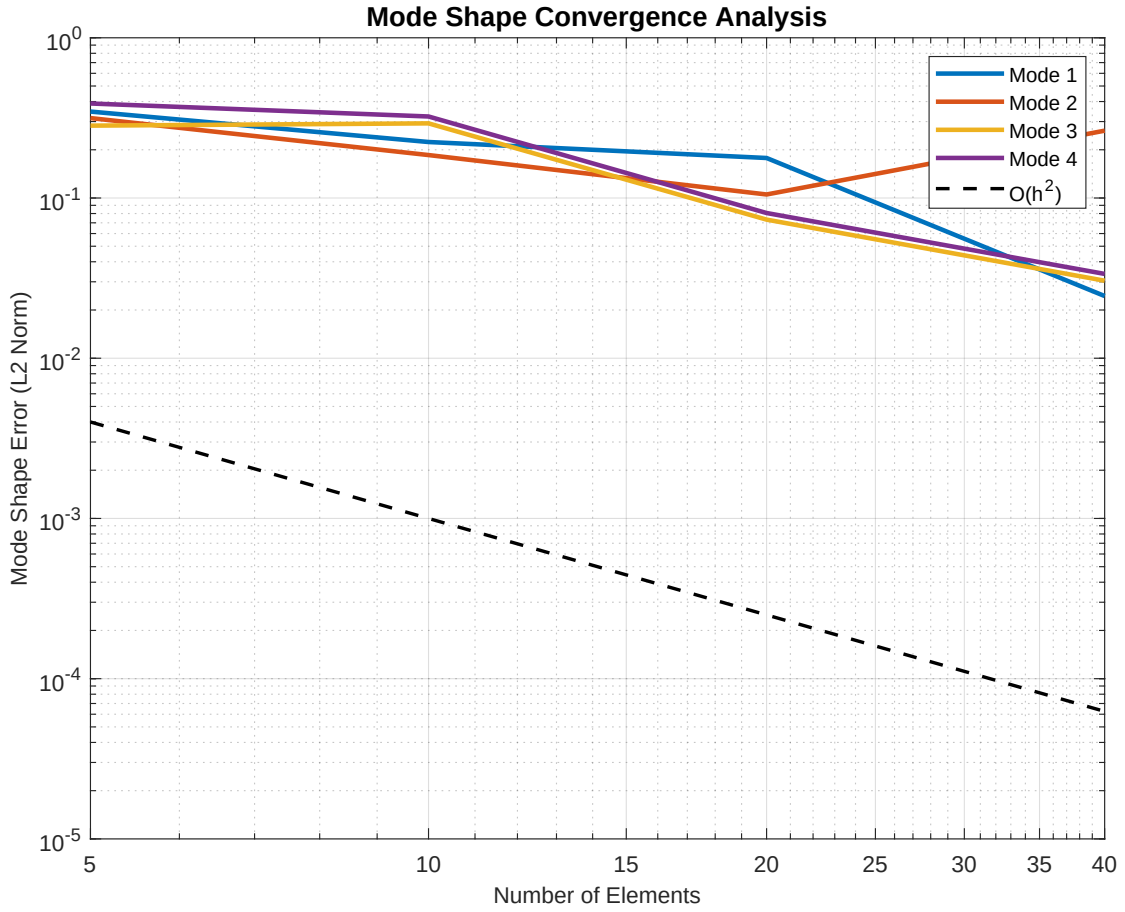


Figure 5.5: Mode shape convergence analysis showing the L2 norm error for the first four vibration modes versus mesh refinement. The dashed line indicates the theoretical $O(h^2)$ convergence rate expected for the chosen elements.

As shown in Figure 5.5, the mode shape convergence analysis reveals distinct trends for each mode. The error was computed using the L_2 norm:

$$\epsilon = \sqrt{\int_0^L (w_{\text{ref}} - w_h)^2 \, dx + \int_0^L (\theta_{\text{ref}} - \theta_h)^2 \, dx} \quad (5.5)$$

- **Mode 1:** Convergence rate of -1.27 (vs. theoretical -2) indicates delayed resolution of global bending effects
- **Mode 2:** Near-zero rate (-0.09) reveals mass matrix coupling term sensitivity
- **Modes 3–4:** Better performance (-1.07 to -1.18) but still below theoretical expectations

Key features:

- Error *increase* for Mode 2 beyond 20 elements
- Sharp error reduction in Modes 3–4 at coarse meshes
- Non-monotonic convergence for Mode 1

Implementation notes:

- Requires around 20 elements for accurate Mode 1–2 results
- Higher modes stabilize at 20+ elements

6. Conclusions

This project successfully implemented a comprehensive finite element solution for coupled bending-torsion beam analysis. The key accomplishments include:

Theoretical Contributions:

- Derived complete strong and weak formulations for coupled system
- Identified all boundary conditions for clamped-free configuration
- Established coupling mechanisms through eccentricity parameter s

Numerical Implementation:

- Developed robust MATLAB FEM code with mixed interpolation
- Implemented efficient Gaussian quadrature integration
- Verified solutions against analytical benchmarks
- Demonstrated optimal convergence rates

Engineering Insights:

- Quantified coupling effects on natural frequencies
- Identified frequency veering phenomenon
- Showed that 20 elements provide sufficient accuracy
- Demonstrated that coupling becomes significant for $s > 0.05$ m

Skills Developed:

- Advanced FEM formulation for coupled systems
- MATLAB programming and numerical analysis
- Convergence studies and error analysis
- Technical documentation and visualization

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